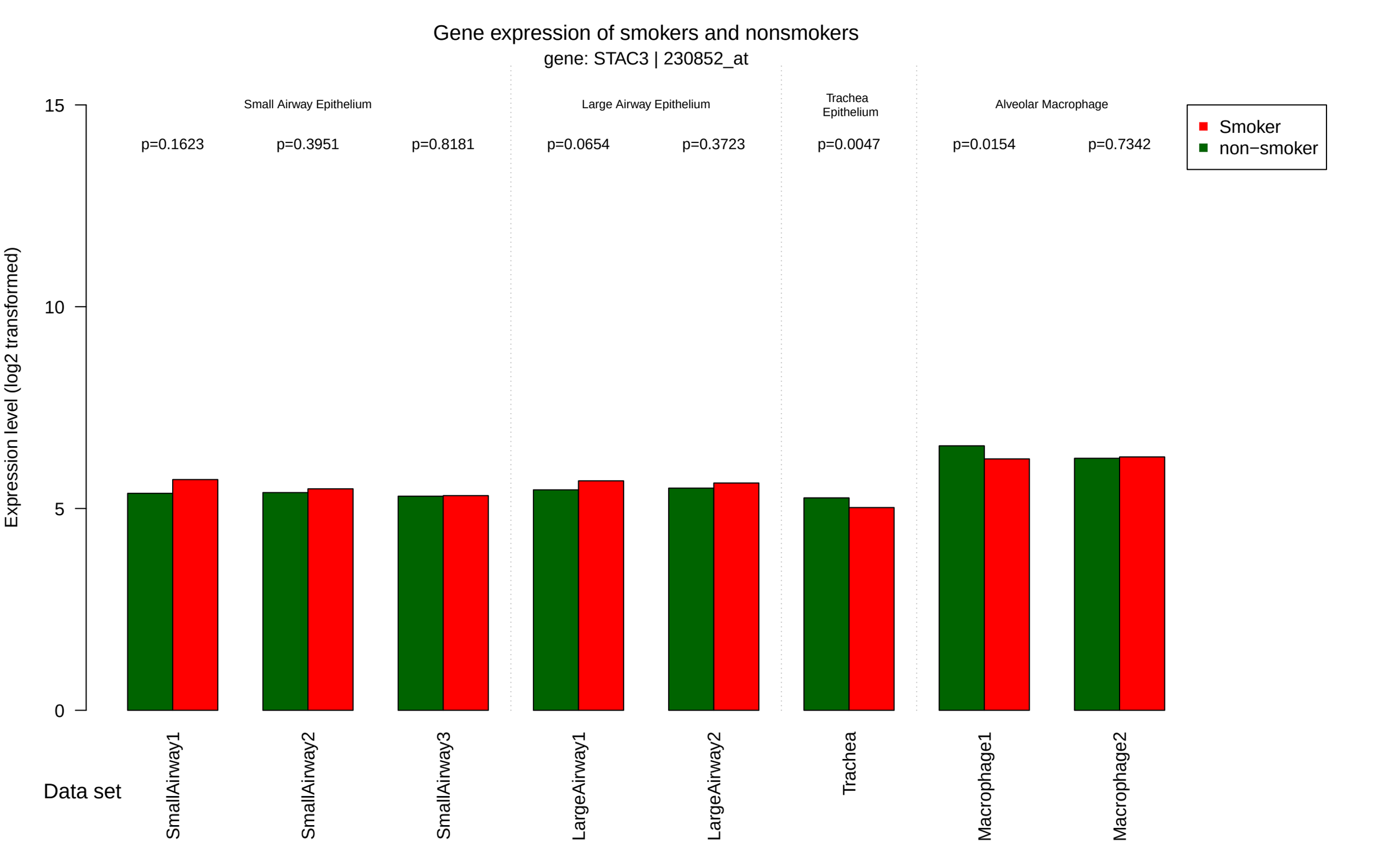
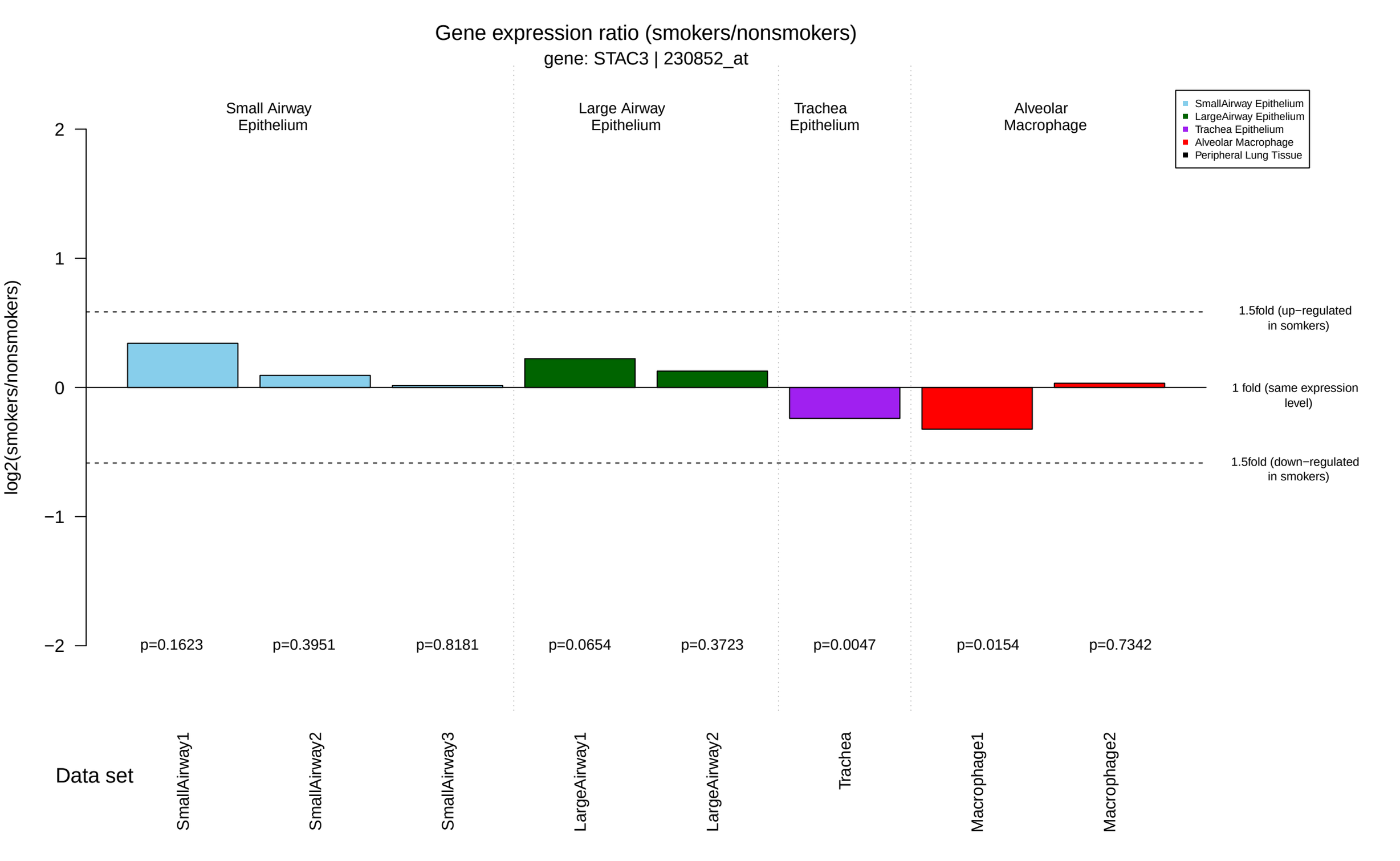
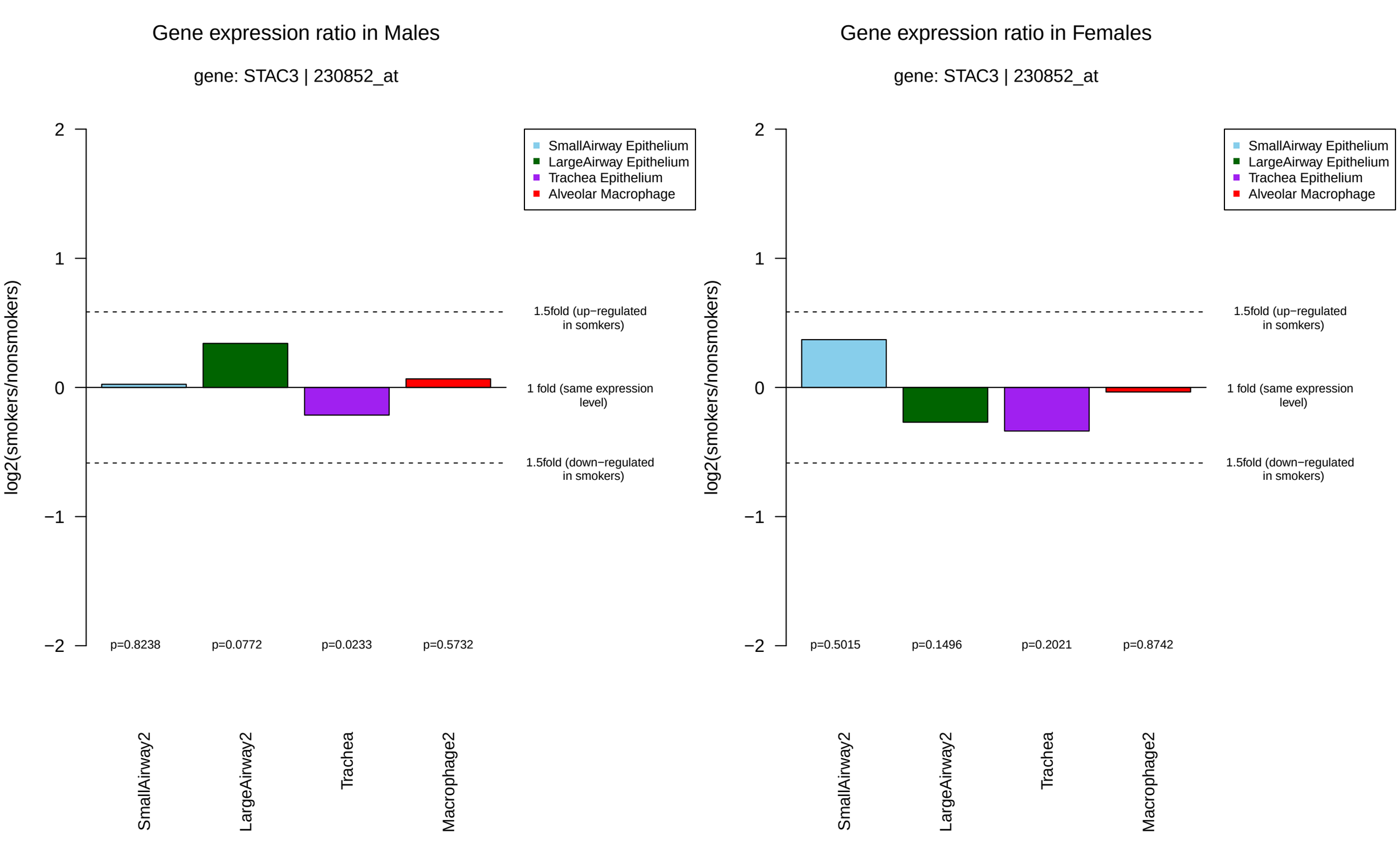
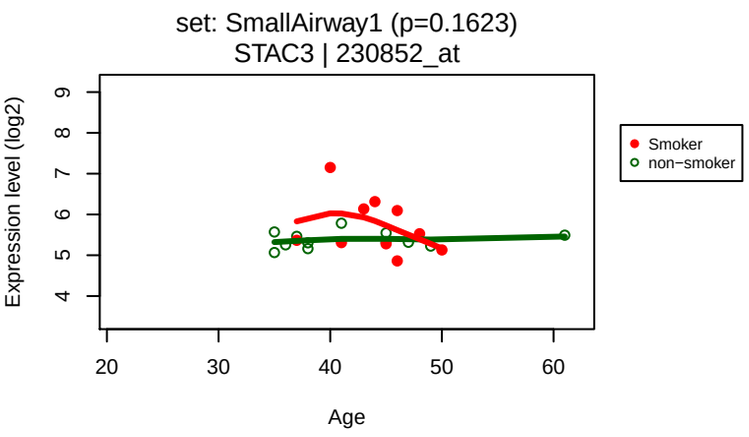
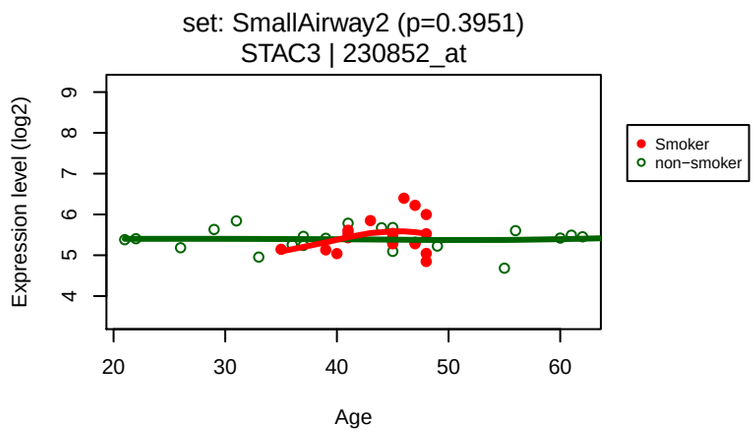
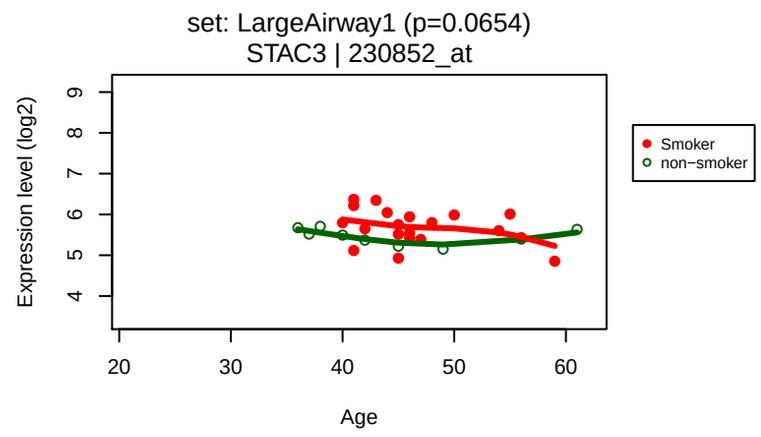

Gene expression of smokers and nonsmokers
gene: STAC3 | 230852_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1623
p=0.3951
p=0.8181
p=0.0654
p=0.3723
p=0.0047
p=0.0154
p=0.7342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

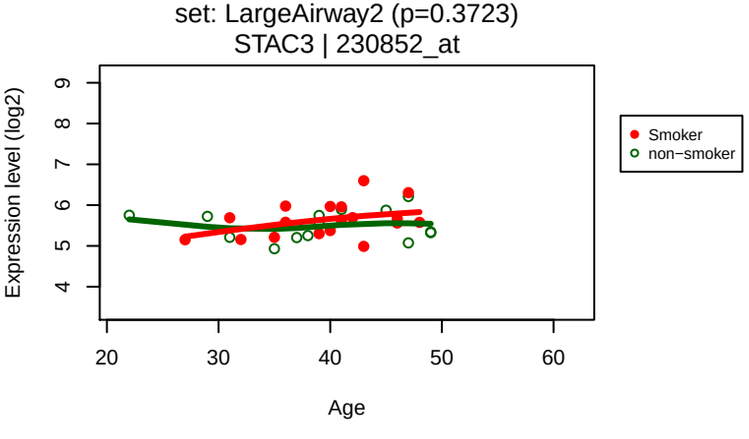

Gene expression ratio (smokers/nonsmokers)
gene: STAC3 | 230852_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1623
p=0.3951
p=0.8181
p=0.0654
p=0.3723
p=0.0047
p=0.0154
p=0.7342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

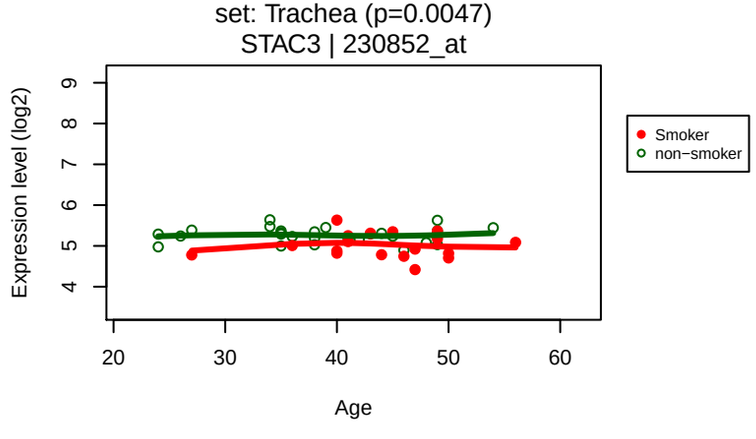

Gene expression ratio in Males
gene: STAC3 | 230852_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8238
p=0.0772
p=0.0233
p=0.5732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAC3 | 230852_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5015
p=0.1496
p=0.2021
p=0.8742
SmallAirway2
LargeAirway2
Trachea
Macrophage2

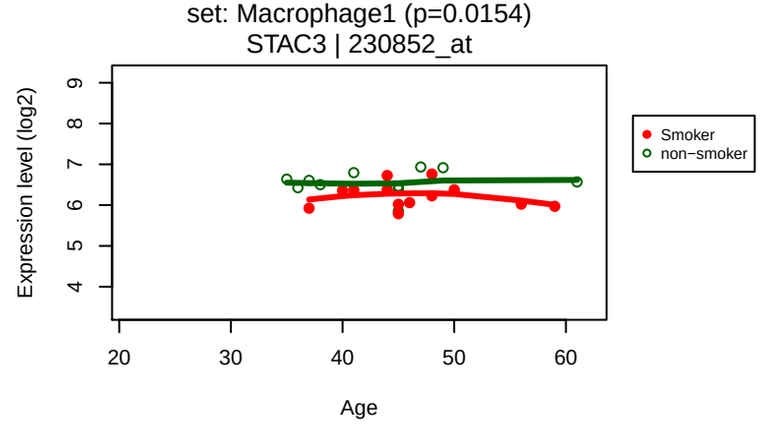

set: SmallAirway1 (p=0.1623)
STAC3 | 230852_at
Expression level (log2)
Age
Smoker
non−smoker

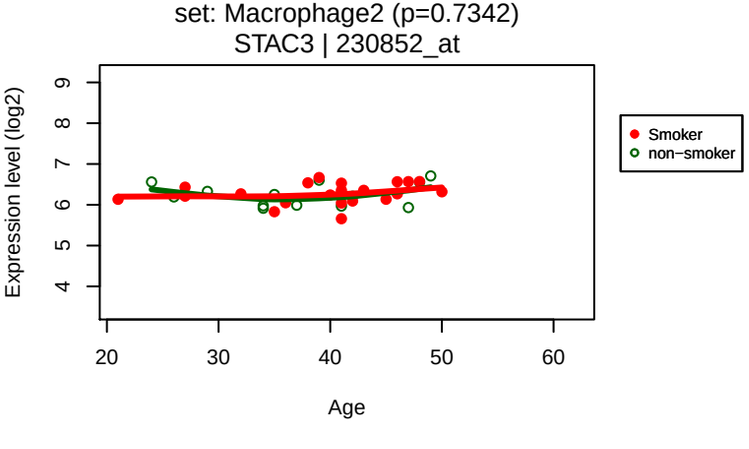

set: SmallAirway2 (p=0.3951)
STAC3 | 230852_at
Expression level (log2)
Age
Smoker
non−smoker

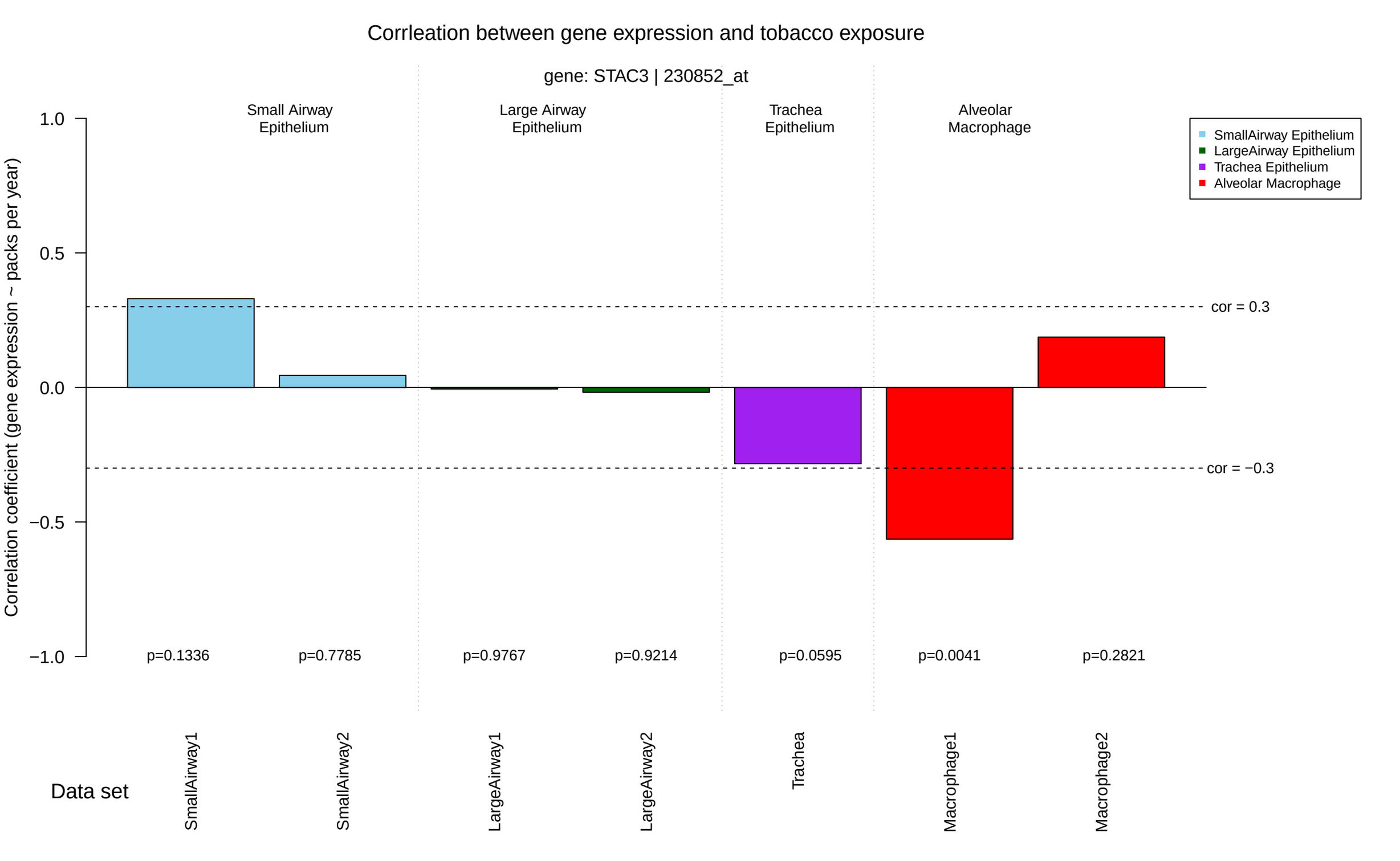

set: LargeAirway1 (p=0.0654)
STAC3 | 230852_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3723)
STAC3 | 230852_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0047)
STAC3 | 230852_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0154)
STAC3 | 230852_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7342)
STAC3 | 230852_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAC3 | 230852_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1336
p=0.7785
p=0.9767
p=0.9214
p=0.0595
p=0.0041
p=0.2821
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2